

# Session\_12

Q.1 Declare a structure named Playlist to store details of a song: title (string), artist (string), and duration in seconds (integer). Initialize one Playlist variable with your favorite song's details and print each field.

A.1 #include <stdio.h>

```
#include <string.h>
```

```
struct Playlist
```

```
{
```

```
    char title[50];
```

```
    char artist[50];
```

```
    int duration;
```

```
};
```

```
int main()
```

```
{
```

```
    struct Playlist song;
```

```
    strcpy(song.title, "Perfect");
```

```
    strcpy(song.artist, "Ed Sheeran");
```

```
    song.duration = 263;
```

```
    printf("Song Title: %s\n", song.title);
```

```
    printf("Artist: %s\n", song.artist);
```

```
    printf("Duration: %d seconds\n", song.duration);
```

```
    return 0;
```

```
}
```

-----

Q.2 Create a structure called FoodItem to store Zomato-style menu data: itemName (string), price (float), and rating (float). Initialize an array of 3 FoodItem variables with real menu items and display their details using a loop.

A.2 #include <stdio.h>

```
#include <string.h>
```

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```
struct FoodItem
{
    char itemName[50];

    float price;
    float rating;
};

int main()
{
    struct FoodItem menu[3];

    strcpy(menu[0].itemName, "Margherita Pizza");
    menu[0].price = 299.00;
    menu[0].rating = 4.5;

    strcpy(menu[1].itemName, "Veg Burger");
    menu[1].price = 149.00;
    menu[1].rating = 4.2;

    strcpy(menu[2].itemName, "Paneer Tikka");
    menu[2].price = 249.00;
    menu[2].rating = 4.6;

    for (int i = 0; i < 3; i++)
    {
        printf("Item: %s\n", menu[i].itemName);
        printf("Price: %.2f\n", menu[i].price);
        printf("Rating: %.1f\n\n", menu[i].rating);
    }
}
```

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```
return 0; }
```

Q.3 Define a nested structure called MovieShow for a BookMyShow-style app: Movie (string), Screen (integer), and a nested structure Time with hours and minutes (integers). Create and initialize a MovieShow variable for any movie and print its details in the format 'Movie: X, Screen: Y, Time: HH:MM'.

```
A.3 #include <stdio.h>
```

```
    #include <string.h>
```

```
struct MovieShow { char Movie[50];
```

```
int Screen;
```

```
struct Time
```

```
{
```

```
int hours;
```

```
int minutes;
```

```
    } time;
```

```
};
```

```
int main()
```

```
{
```

```
struct MovieShow show;
```

```
strcpy(show.Movie, "Avengers");
```

```
show.Screen = 3;
```

```
show.time.hours = 18;
```

```
show.time.minutes = 30;
```

```
printf("Movie: %s, Screen: %d, Time: %02d:%02d\n",
```

```
show.Movie,
```

```
show.Screen,
```

```
show.time.hours,
```

```
show.time.minutes);
```

```
return 0;
```

```
}
```

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---

Q.4 Build a structure called InstaProfile with fields: username (string), followers (integer), and a nested structure Bio with fields: description (string) and age (integer). Initialize an InstaProfile variable with your own details and display all fields.

A.4 #include <stdio.h>

#include <string.h>

struct InstaProfile

{

    char username[50];

    int followers;

    struct Bio

    {

        char description[100];

        int age;

    } bio;

};

int main()

{

    struct InstaProfile profile

    strcpy(profile.username, "kittu");

    profile.followers = 1500;

    strcpy(profile.bio.description, "C programming student");

    profile.bio.age = 20;

    printf("Username: %s\n", profile.username);

    printf("Followers: %d\n", profile.followers);

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```
printf("Bio: %s\n", profile.bio.description);  
printf("Age: %d\n", profile.bio.age);  
return 0;  
}
```